

SECTION TWO

The Methodological
Basis of Endurance



CHAPTER THREE

The Methodologies of Endurance Training

In the previous section we explained the basics of what enables endurance and the physiology that supports it. To establish the knowledge for this endeavor, we mentioned early in Section One that we need to ensure we're all speaking the same language. In this section we will establish a common vocabulary of terms and principles and then we'll investigate how these principles improve endurance.

A Little History

What are the methods used to increase endurance? Where do they come from? Why are they successful? Just as with the physiology of endurance, there is an intellectual framework that underlies the whole methodology of endurance training. The better handle you have on this framework, the better you can understand and apply the correct training for your situation.

Over the past one hundred years, a vast body of knowledge concerning the body's response to various forms of endurance training has been amassed. Despite the prodigious amount of scientific research examining the myriad qualities that comprise endurance, it is the coaches and not the scientists who have been at the forefront of establishing the most effective training methods. Through a trial-and-error approach, the coaches of the past have discovered what did and what didn't work. Failed training methods were discarded or modified until the most effective methods rose to the top in an evolution-like process. The scientists come along later to shed light on why those successful coaching methods work, and that in turn supports the intellectual framework. Having an understanding of some of the concepts involved will create some context for your own training decisions.

Previous page: Alyson Dimmitt Gnam and Katie Rogotzke running ridgelines in the Two Medicine region of Glacier National Park, Montana, while a wildfire flares up in the distance. Photo: Steven Gnam

Opposite: Kilian Jornet doing his aerobic capacity training, Romsdal, Norway. Photo: Sebastien Montaz-Rosset

NOTES FROM KÍLIAN

As a result of the combination of my education in exercise physiology and working with coaches; especially as a junior, I learned what the expected result would be from every training action. What an easy run means for my fitness; what a fast-tempo run means for my fitness. And, perhaps most importantly, how it all fits together. How to build the base broad before doing faster, harder work. With both book learning and learning from my coaches, I built up the knowledge that allows me to understand the expected effect on my body that will result from every training day. I understand why I can go long easily. Why I recover well. I've been a student of myself, my own reaction to training, as well as the

science.

I also studied a lot about running mechanics during my school years—I really worked hard on my running technique. How to be relaxed. And when I notice I'm not efficient, I can figure out where the problem is coming from and try to fix it.

And then there is stress. I believe it is very important to appreciate how nice it is to be out running in the summers and skiing in the winters. I think it is very important that training is not stressful in and of itself. It should be fun, enjoyable, with good people, and in nice places. This reduction in stress is important to recovery, and therefore progress.

Become a Student of the Sport

Doing an experiment on a sample size of one (yourself) is risky and likely to be misleading. Self-coached athletes owe it to themselves to learn as much as they can about not only what methods do and don't work, but also why. Avail yourself of this knowledge base and hopefully you can avoid making unnecessary mistakes with your training. Rest assured that it has all been tried before.

Important Concepts, Terms, and Principles

Building the knowledge base with which to view training programs, and even individual workouts, allows you to think in terms of cause and effect. It gives a perspective that moves training from blindly following a recipe to making informed decisions. A great deal of coaching (especially of yourself) comes down to artfully applying certain time-tested methods. As with other sciences, our understanding of complex systems is enhanced through the use of simplified models. While we recognize the limitations of models for their predictive ability, they are still useful for understanding systems as complex as the human body and its responses to training.

The Training Effect

“Training is not the work you do but the effect it has on your body.”

– Renato Canova

When some of the physiological systems of the body move too far outside their homeostasis (comfort zone), those systems are put into a crisis state. When this destabilization is caused by a training stimulus, it triggers the start of what we have come to know as the Training Effect (see glossary). The stress of training causes certain adaptive responses to occur: first manifested as fatigue and later recognized as adaptations that enable a greater resistance to that same stress when applied later.

The concept of the Training Effect was a consequential offshoot of Hans Selye’s Generalized Adaptation Syndrome. Selye, a Viennese medical doctor and chemist, coined the term *stress* as we know it today in the biological context. His theory explained, in a coherent manner, how biological stressors acted upon organisms in both beneficial and detrimental ways.

The following presents the cascade of effects occurring during and after the application of training stressors or Training Stimuli (glossary), as it is often referred to.

Examples of the Training Effect

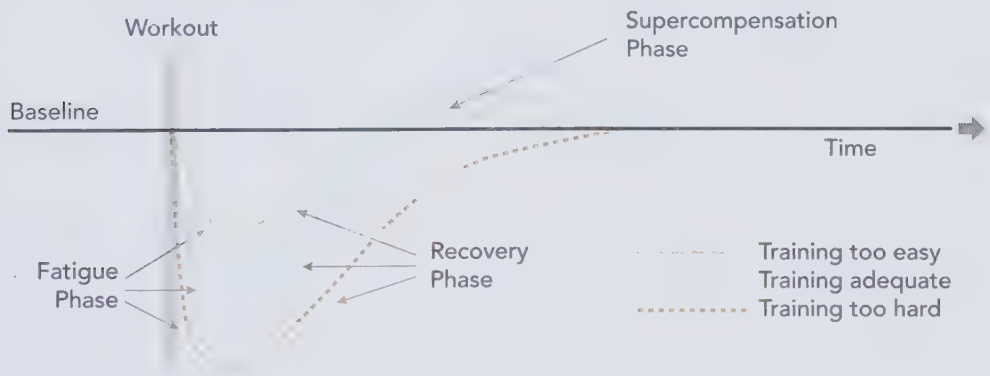


Fig. 3.1 Examples of the Training Effect

The result of a training session is fatigue and a temporarily diminished work capacity. During the recovery phase is when the adaptations occur that increase a person’s fitness level. Applying a new training stimulus during the supercompensation phase will over weeks and months result in improved fitness. Too frequent training stimuli with insufficient recovery or too infrequently applied training stimuli with too much inactivity will both result in the adaptation process not occurring and the athlete will become less fit.

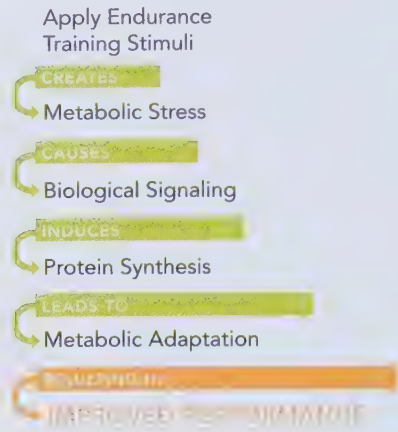


Fig. 3.2 The Endurance Training Cascade
Endurance training stimuli cause stress that triggers a cascade of events leading to improved performance. These changes go right down to the gene level. Genes are either enhanced or moderated by training stimuli.

Capacity Versus Utilization Training

Imperfect as it may be, one of the more helpful models of training is the distinction made between Capacity Training and Utilization Training. Understanding whether a workout’s goal is to increase your capacity for that sort of work or to maximally utilize whatever capacity you bring to the workout can help with placing individual workouts into a plan.

Among endurance sports, USA Swimming has had the most consistent and unparalleled international success for decades. This success has caused the athletic and coaching world to take notice. Bob Bowman, most famous as Michael Phelps’ coach for his entire phenomenal career, has used the terms Capacity Training and Utilization Training to describe two different types of training that swimmers can use. Here is how Bowman defines these terms:

- **Capacity Training:** Training that improves the *long-term* performance *potential* of the athlete. Capacity Training is commonly prioritized during the Base Period. This training acts to improve the fundamental qualities needed to support participation in the event itself and Utilization Training. As such, it is often not sport specific.
- **Utilization Training:** Training that improves the *near-term* performance *results* of the athlete. Utilization Training is commonly prioritized during the build-up to the competition period or the targeted event. This training models the specific demands of the event an athlete is training for.



Racers climb above the Col du Tourmalet, on the French side of the Pyrenees, during the 2018 Altito-Ternua race, the biggest and most popular skimo race in Spain. Photo: Davide Ferrari

A debate has raged in sports for decades: Is it more effective to focus the training of an athlete on the more general qualities (increasing Capacity) with a short Utilization phase before competition? Or is it better to train primarily by doing event-specific workouts that maximally utilize whatever capacity the athlete currently has?

The sport of running in the US dealt with this debate at the turn of the century after the United States' success of the 1970s and 1980s gave way to an era of disappointing results in the 1990s and early 2000s. The later era coincided with a shift by many coaches and athletes away from a capacity-oriented training system to one relying heavily on a utilization approach. Rowing had this debate back in the 1980s after a dramatic change in the coaching philosophy in Germany which abandoned the utilization approach in favor of one based on building capacity. This switch soon led to German domination of the sport. Cross-country skiing had a similar internal debate when, in 2004, Norwegian exercise physiologist Dr. Jan Helgarud caused a sensation with his outright denunciation of Capacity Training as useless.

These two training models have been well tested by millions of athletes and thousands of coaches in the ultimate laboratory: the competitive arena. While some noteworthy holdouts exist, the approach of relying primarily upon building capacity before applying limited amounts of Utilization Training has largely won out in



Colin Haley and Sébastien Montaz-Rosset ascending the Argentière Glacier near Chamonix, France. Colin and Ben Tibbetts, supported by Sébastien and Vivian Bruchez, made the first north-south traverse of the Mont Blanc massif: 6,800 meters of climbing and 89 kilometers of ski mountaineering in 32 hours. Viewed as training, this would be considered Utilization Training. Viewed as an objective, this is a huge goal. Photo: Ben Tibbetts

this contest of ideas. Kilian's legendary ability to handle a huge volume of work (and much of it very hard) is a result of his decades of capacity-building training. It can be very tempting to try to simulate the training of elite athletes. Doing so without their years of capacity-building work will mean that you are actually doing Utilization Training. As you read on, you will see why this is not the best path to long-term gains.

Different sports use different terms to describe these concepts, but Bowman's is perhaps the simplest to understand. This is how he explains the terms.

CAPACITY TRAINING

Capacity Training increases the ability to do work in each of the various realms required for the sport. In uphill athletics (mountaineering, mountain running, ski mountaineering, and skimo), here is what we'd call Capacity Training:

- **Aerobic Capacity:** This training improves the main locomotive muscles' ability to produce ATP via aerobic metabolism. This results from increases in mitochondrial mass, capillary density, aerobic enzymes, and cardiac output. This should be the top priority for all endurance athletes. An endurance athlete can never have so much aerobic capacity that it is detrimental to performance.

- **Anaerobic Capacity:** Increases the ability of the glycolytic metabolism to produce ATP. An endurance athlete *can* have too much anaerobic capacity and this depends very much upon the event being trained for.
- **Strength Capacity:** Increases the maximum muscular force in sport-specific movements along with local muscular endurance in the main locomotive muscle groups. Greater strength is beneficial up to a point. Excessive strength can cause problems for mountain endurance athletes when they add significant muscle mass. We will discuss this in detail in the section on strength.
- **Technical Capacity:** Improves the economy (energy cost) of locomotion, balance, and proper movement patterning.

Notes on Capacity Training:

- Its gains in general take a long time to accrue. Its effects tend to be longer lasting than those brought about by Utilization Training.
- It usually involves training each of the above-mentioned qualities in isolation. As such, it will appear as less sport specific and more general in nature; it may not have a direct relation to the competition or event being trained for.
- It is meant to improve the performance *potential* sometime in the future and may even reduce near-term performance. It's analogous to building the Interstate highway system: Tedious and dull during the construction process and may cause a slowdown in traffic but once completed, it allows much more traffic to flow (work to be done) more quickly (utilizing one's capacity).

UTILIZATION TRAINING

Utilization Training is used to increase performance *outcomes* in the short term. For a runner or a competitive ski mountaineer, this type of training would be racing or race-specific workouts. These will maximally utilize whatever capacities the athlete currently has. Extending the Interstate analogy: It dumps a lot of fast-moving traffic on that Interstate system. If the highway system can handle the increased traffic flow, then a lot of work gets done very fast. If the highway system is still under construction with restricted speed limits, then no matter how much traffic hits the highway, the amount of work being done will be severely limited by the capacity of the roads to handle the traffic. Utilization Training is quicker acting than Capacity Training. For maximum effect, it should be sport/event specific. It is usually of an intensity that mimics the event being trained for.

Here's how Utilization Training looks by physiological system:

- **Aerobic Utilization training** is also called endurance training, LT training, or VO_2 max training (Zones 3, 4, and 5 in a conventional zone system). It increases the athlete's maximum sustained output. While it plays a role in all endurance

training programs, this high-intensity training is especially important for events with durations under two hours where the maximum competition speeds are at and above the LT. For long-duration events, over three hours, the Aerobic Capacity and Aerobic Utilization Training are essentially the same because most of the event will be competed at speeds below the LT and nearer to the AeT.

- Anaerobic Utilization increases the fraction of the Anaerobic Capacity that can be sustained in competition. It typically involves training at speeds 5–10 percent faster than competition speeds or under a workload significantly greater than those encountered in the event for a total duration significantly less than the competition distance. This type of work has a strong training effect on muscular endurance capacity.

Notes on Utilization Training:

- Utilization Training will reduce the athlete's capacity and this effect must be offset with Capacity Training *during all phases of the training cycle*.
- The effects of Utilization Training are dynamic and volatile. While rapid gains are seen when implementing Utilization Training, they are often short lived. If you've had back-to-back awesome and horrible races, or find it difficult to string together a full season of good races, one potential cause is too much Utilization Training as compared to Capacity Training.
- Utilization Training will be necessary to enable an athlete to achieve his or her personal best. However, Utilization Training cannot be expected to have its optimal effect if the athlete's capacity is small. Think again of the Interstate highway system.
- Athletes with a large capacity can and must do much more Utilization Training to maximize results. This explains why elite athletes do more and harder training than do novices, who have less capacity. It also explains one of the most common mistakes made in training, which is for amateurs to copy the elites.

It's clear that Capacity Training and Utilization Training both have their place, and are interdependent. But it is Capacity Training that takes the majority of an athlete's time. Only when the capacity for the sort of work needed by your event is sufficient will you maximally benefit from adding Utilization Training. The result of Utilization Training, which can be astonishingly quick and seductively satisfying, is the reason that any endurance athlete is wise to take the long view. Real endurance is built from year to year, not weekly or monthly.

Next page: Katie Bono laying down some aerobic capacity training while running the Pinnacle Peak Trail in the Tatoosh Range with Mount Rainier, Washington, in the background. Katie holds the female speed record on both Tahoma and Denali. Photo: Jason Thompson

Understanding the Language of Intensity

I monitor intensity with a heart rate monitor. However, I've been doing it so long I don't need a watch anymore; I can tell you my heart rate within a few beats per minute, simply by feel.

– Kilian Jornet

Determining the appropriate intensity for training is of the utmost importance in all endurance sports. This intensity dictates the energy systems and how many and which type of muscle fibers (ST or FT) get recruited to do the work—the effect of the training is directly related to the intensity. The goal of this discussion is to allow you to target your training to achieve the intended outcome.

Intensity can be measured with power meters in cycling and pace in road and track running. Uphill athletes train and race on a variety of terrain and conditions that makes heart rate (HR) the only practical method of controlling intensity.

A Brief Review of the Metabolic Markers of Intensity

Aerobic Threshold (AeT)

When ventilation is slow and controlled—you can carry on a conversation without needing to catch your breath or can breath through the nose only—aerobic metabolism is providing the majority of the energy and this intensity can be maintained for hours. This is referred to as the AeT or First Ventilatory Threshold (VT1). The rate of breathing makes a jump above this point. In the majority of endurance-trained athletes, the AeT corresponds to the point where the blood lactate concentration has risen modestly (1mMol/L) above a baseline reading to a value of about 2mMol/L.

Lactate Threshold (LT)

As the speed (intensity) increases, so does the body's energy demands. Eventually the lactate removal rate can't keep pace with the lactate production rate. This point is normally referred to as the LT. It represents the maximum intensity at

which lactate levels will remain elevated but stable for up to an hour at a time. Above this intensity, lactate levels climb quickly and the athlete will eventually be forced to slow down.

Every endurance athlete knows this point. Cross it, and you know you can hold that pace for only a short time before you have to dramatically slow or stop. The LT is the best proxy for endurance. Raise it (either speed or heart rate or both) and you improve your endurance performance. Luckily, it is the most trainable of all endurance qualities.

This point corresponds well with a Second Ventilatory Threshold (VT2) where breathing suddenly becomes noticeably deeper and more rapid. Above this, conversation becomes impossible; only a few words can be spoken between breaths. Above the LT, exercise duration will be measured in seconds to minutes. The higher the intensity, the shorter the time.





ATHLETE STORIES

A Long Road to the Same Place I Started

Dakota Jones

For my first adventure I set out to traverse a range of peaks that stood above my home. At the age of seventeen, I had just discovered that there are whole groups of people who spend their free time going as far and as fast in the mountains as possible. Recently I had come to the surprising conclusion that there was very little I'd rather do than explore wild places on foot. So, I set out on my adventure with a backpack containing some food, some water, and little else.

Four hours into the climb, I emerged from the pathless forest onto a high ridgeline dividing two major drainage basins. Ahead of me stretched the rocky spine of mountains, their sharp summits outlined against the brilliant blue sky. There were four peaks to climb, and the descent lay on their far side.

Not long after, I sat at the base of a steep wall of rock that had stymied my best efforts to climb it. Without rope, experience, or partners, I decided to descend, but not by the way I had climbed, which from this vantage seemed roundabout and overlong. Instead I hiked down into the basin below me, knowing it would drain down to the river I had crossed in the morning. In this way I embarked on my first epic.

The basin below me was calm and still, the lake reflecting the darkening clouds above. Farther on, the creek dropped off in a 300-foot waterfall, and I slid down the cliffs alongside by hanging from tree trunks clinging to the hillside and then dropping to the next one. Below that I marched through a nearly flat valley grown over with bushes taller than me and crisscrossed by invisible logs at knee level. Escaping to the grassy hillsides, I encountered swarms of black-flies that clung to every inch of exposed skin, biting painfully. I was forced to sprint along the hill, jumping between rock outcroppings like a mountain sheep. Lower down, while I was walking knee-deep in the

creek, the canyon walls narrowed and became vertical, forcing me to swim through long stretches of icy water. When I reached my car after eighteen hours, dark had long since fallen and I was exhausted, dirty, scratched, and ravenously hungry. But I'd emerged, and my despair instantly transformed to elation.

Many adventures followed, until I became so accustomed to mountain travel that the word *adventure* quit being relevant. Real adventures involve uncertainty, fear, struggle, and the euphoria of overcoming them. But as I pursued a passion for movement and exploration, I got really good at running long distances—good enough to become a professional trail runner. The desire for running success started taking me out of the mountains, as I pursued specific workouts like intervals, at specific elevations, in conjunction with specific food and cross-training routines. Athletes must learn to schedule their lives so as to maximize their time doing beneficial activities (like training, eating, sleeping, etc.) and minimize time doing nonessential things (like having friends, working, and going on adventures). I became a mountain runner who spent very little time in the mountains.

My single-minded focus on running success eventually led me to injury. With a “more is better” attitude, I relied on my physical strength to carry me farther than anyone else, with the result that I played myself out in training and suffered in races. I developed a stress fracture in one foot, and after that healed I strained a hamstring, and while recovering from that I pulled my Achilles tendon. This took place over a matter of years. My career, such as it was, began to fall apart because I simply couldn't do the thing I was best known for. When I was able to train, I returned to the patterns that had injured me before, desperate for a comeback. The mountains were always in the near distance, but adventure had long since taken on a different meaning.



Dakota Jones having a great time running up La Tournette above Annecy, France. Photo: Martina Valmassoi

As you read this book, keep in mind that training to be an uphill athlete is simply a mechanism to make you more able to do what you love. Strength and preparedness will give you the tools to take advantage of opportunities, and in turn you have to remember to continue to take opportunities. It won't help you to be the greatest uphill athlete if you forget to go to the mountains and actually go uphill. I spent so much time training to be a great runner that I never got to actually run, and though my life is centered on mountain adventures, those adventures have been few and far between in recent years. The passion we all feel for exploration in wild places must be the guiding light that directs us through the confusing milieu of culture and sport. Training is important. But it is only a means of facilitating what you love. Don't forget what you are training for.

These days I'm trying to strip away some of the certainty in my life and go back to the attitude that sent

me on an eighteen-hour hike/climb with little food and no map. I want to remember to laugh and keep moving even when I can't see the ground in front of me. I want to remember what it's like to truly have no idea what might be around the next corner, and if I'm going to make it home in time for dinner. If I train well—not more, but smarter—then I'll be prepared to deal with the unexpected, and my adventures can be adventurous again, and maybe even a little fun. There can be struggle in enjoyment, if you're willing to try something new. And there can be happiness in uncertainty, if you train for it right. At the end of the day, the point is to have a good time. ♦

Dakota Jones is a runner from Durango, Colorado, who's currently having a great time.

A Discussion of Intensity Zones

Each of the intensity HR zones described here has a specific role to play in the overall scheme of the training for any athlete in any endurance sport. The anchors for this five-zone system are the two metabolic markers we covered in the Physiology Section, AeT and LT, and repeated in the sidebar nearby. These two markers are highly trainable, so these points are not fixed and you can expect to see your zones shift a small amount on a long-term basis due to fitness changes. By basing the scale primarily on these personal metabolic events, you will have a personalized zone distribution offering much more accuracy than those based off percentages of max heart rate. Methods of determining these two markers are addressed in Chapter Five.

Let's be clear that your metabolic response to exercise does not fit neatly into discreet zones with hard boundaries as is implied by the zone system and chart below. Not only is there a great deal of variation between individuals, there is some variation from day to day for each athlete. A heart rate of 145 on a day when you are fully recovered is going to have a different training effect on the day after a hard effort. As with most of our attempts to understand the very complex human body, the zone system is a flawed model. However, it is a model that can still shed useful light on the control of intensity. Keep in mind that there is no single agreed-upon zone system. Systems range from three zones delineated only by the AeT and LT metabolic markers up to a seven-zone system that

Intensity Zone Systems

There is understandably a fair bit of confusion surrounding the several different zone systems in use. Let's start by saying that all of them are flawed to some extent. All zone systems attempt to associate heart rate with intensity. The body's response to exercise will vary from day to day depending on a number of factors—recovery state being the most powerful one affecting where an athlete's zones lie on a given day. This means that the principle metabolic markers that anchor the zones may move around by several beats per minute from day to day. Expecting the same heart rate to represent the same physiological state during every workout is a big assumption. This also should cause you to question the parsing of zones into subzones that purport to define

intensity zones to within one bpm; this is what scientists call false precision. Not only are the factors just mentioned at play, but it is impossible to control the heart rate with that level of precision in real-world training. When you read about a seven-zone system, carefully consider whether it can actually aid you in your training.

We suggest using the traditional zone system described nearby, which is anchored by the two primary aerobic metabolic markers: AeT (top of Z2) and the LT (top of Z3). This gives the broad separation of the aerobic zones (Z1 and Z2) for easy to moderate training from the anaerobic zones (Z4 and Z5) where training is very hard and of shorter duration. This leaves Z3 in the middle as hard training.

	Heart Rate	Perceived Effort	Training Effect/ Purpose	Metabolism	Muscle Fiber Recruitment	Training Method
Zone 5	N/A.	Supra max, unsustainable, exhausting.	Power, speed, technique, anaerobic capacity, anaerobic endurance.	ATP/CP and glycolytic, minimal aerobic involvement.	All ST + all FT.	Interval 8-60 sec.
Zone 4.	Lactate Threshold to LT to maxHR.	Hard, max sustainable.	Maximal aerobic power, strength/speed endurance, economy, technique	Both aerobic and anaerobic capacities maxed out	All ST + most FT.	Interval 30 sec-8 min.
Zone 3	AeT to Lactate Threshold.	Medium, fun-hard not exhausting.	Aerobic capacity, anaerobic capacity, lactate shuttle, economy.	Glycolytic/anaerobic begins to dominate.	All ST + some FT.	Interval 10-20min, continuous to 60min
Zone 2	AeT-10% to AeT.	Moderate for those with high AeT, easy for those with low AeT.	Aerobic capacity, economy.	Aerobic-fat dominates, maximum fat utilization.	most ST.	Continuous 30-90 min.
Zone 1	AeT-20% to AeT-10%.	Very easy to easy.	Aerobic conditioning.	Aerobic-fat.	ST.	Continuous 30 min to several hours.
Recovery	Under AeT-20%.	Light.	Recovery.	Aerobic-fat.	ST.	Continuous 20-60 min.

Fig. 3.3 The Five Training Zones

This chart shows a zone system using the AeT and LT as the significant metabolic events as anchors. It also lists the training effects, the metabolic process, and muscle fibers that dominate each zone and the training method used to improve this quality. The percentage ranges above and below the AeT and LT are for guidance only and should not be treated as hard boundaries. Only careful lab testing will determine these points. The Recovery Zone is not clearly defined. Its boundary with Zone 1 is indistinct depending on the recovery state of the athlete. Hence it does not get its own numeric zone designation.

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ATHLETE STORIES

Training for Mountain Running Without Mountains or Time

John Kelly

If you're reading this, then chances are that you, like me, would love to live in the mountains, or maybe you already do. I often daydream of escaping, staring longingly at the slideshow of pictures that plays in our living room, hanging where most people would have a TV. The solitude, the serenity, the scenery . . . not to mention the incredible training I could get in. But most of us can't drop everything and move to Montana.

My reality is that I live just outside of Washington, DC, with my wife and three kids, in a congested area where finding a hill with 100 feet of gain is a serious challenge. I also work a demanding job at a tech startup, leaving little time to escape to the mountains.

Most people in the ultrarunning world know me from being the winner and sole finisher of the 2017 Barkley Marathons, a race that boasted over 67,000 feet of elevation gain—more than climbing Everest twice from sea level. Without the proper terrain or time, how did I train for a race like that? Admittedly, not as effectively or enjoyably as I could have if I lived in Montana, but the results prove that it is possible.

My “letter of condolence” accepting me to the 2015 Barkley Marathons was oddly appropriate. The race had long appealed to me and held enormous personal meaning—my family has lived on the land alongside the course for more than 200 years—but to be honest I wasn't expecting to get in. The thought of finishing Barkley was like the thought I had as a kid of striking out the last batter with the bases loaded to win the World Series. But it was fun to dream about.

So there I was with my condolences, absolutely no idea what I was doing, and a nice conversation to have with my wife that started, “Yeah, so remember that thing I said I'd apply for but that almost definitely wouldn't happen?”

I had never trained for anything even remotely resembling Barkley. In my head I frantically went through all the nearby hills. There was one that went up 90 feet in half a mile, or a little slope that went up about 15 feet over 130 feet. I got on Google Earth and scanned everything around for the tightest contour lines I could find. I zeroed in on Mar Lu Ridge: it was about an hour away but had a much more appealing 500 feet up in half a mile.

So I ran them. I ran all the hills. I had no real plan, no real purpose. My mind, overwhelmed with the enormity of Barkley, latched on to the one certainty it could: must run hills. On weekday evenings, I ran the nearby hills. Weekends involved pre-dawn drives to Mar Lu Ridge, with my biggest workout going all the way past sunset: over 40 miles with about 20,000 feet of gain in single digits with a foot of snow. *Perfect miserable conditions for Barkley training!* I thought.

But in the end I still felt unprepared. And worse, it had consumed my life: evenings and weekends were spent training while my wife was home with our infant son. I managed to complete a Fun Run (three out of five loops) that year, which was an incredibly surprising result to nearly everyone. And it was actually a poor nutrition plan rather than physical training that did me in; turns out that you need a bit more than gels and energy bars for thirty-plus hours.

After that experience I knew better what I needed to do, and I knew that I could do it. In 2016 and 2017, my training focused more on time efficiency and effectiveness rather than just volume and elevation. I found a new hill nearby—95 feet in 1/20 of a mile—and used it on the weekends. I no longer lost two hours just driving.

John Kelly, snatching a page from one of the books, before becoming the winner of the Barkley Marathons, 2017, Frozen Head State Park, Tennessee. Photo: Alexis Berg





John Kelly touches the yellow gate, finishing the Barkley Marathons and becoming one of the few finishers in the race's history. John is wearing a beanie he found on the course and the remains of a garbage bag, which he used as a makeshift rain jacket. Photo: Josh Patton

On weekdays all my running was done as my commute—free in my time budget. There wasn't much elevation, but I focused on intensity to maintain the LT needed to survive sixty hours of Barkley hills. I added incline treadmill workouts on weekends so that I could be at home to free up my wife while the kids slept or played. None of these runs were as great as stepping out my back door into the mountains, but it's what I could do consistently and it allowed for more time for family.

By shifting my focus to efficiency, I actually improved the quantity by creating more time for it: my 2017 Barkley training was about 20 miles and 6,000 feet of gain more per week than in 2015. A lot of that gain was on the treadmill, which meant less of the critical but often overlooked downhill running and building of trail muscles. But otherwise my miles were higher quality, with structured workouts tailored toward Barkley's demands.

Training is an optimization problem with a complex set of constraints, unique to everyone. Very few people can spend the optimal amount of time training for mountain ultramarathons: after things like work,

family, and sleep, there simply aren't enough hours left in the day. The best anyone can do, in training or for the most part, life in general, is to work with the conditions given. But that doesn't mean all of us with busy lives and no mountains must abandon hope; it just makes for a puzzle to solve to fit in the best-quality training possible however we can. And then like Barkley itself, when success does come, it's all the more satisfying knowing the obstacles that were overcome to achieve it. ■

John Kelly is an ultrarunning data scientist with a triathlon habit. He has won the Barkley Marathons, finished in the top ten in the men's 30-34 division at the Kona Ironman, and holds the world record for fastest marathon dressed as a video game character. He also holds a PhD from Carnegie Mellon University and works as the director of analytics at QxBranch. This proud father of three (including twins!) resides in Rockville, Maryland. Follow John's running saga at randomforestrunner.com.

seems to split metabolic hairs with false precision. We're presenting here the most common five-zone system. Though other zone systems exist, we have found the time-tested five-zone system to be the most useful model to address real-life training decisions. Below we take a more detailed look at what is described in the five-zone chart.

RECOVERY ZONE

Below Zone 1 lies an intensity that plays a minor role in improving aerobic fitness. But, it plays a major supporting role in that it speeds recovery from the more demanding training sessions. You should feel better right after or within a few hours of completing one of these. It can be almost any form of light exercise; you may benefit most from a modality different than your primary one. One of our favorites for hammered legs is a swim. No specific duration is given in the accompanying charts because this is so individual. For some, a fifteen-minute walk before bedtime will be perfect. Some people will use daily foam rolls or yoga. Others may find a one-hour slow jog does the trick. Heart rate for recovery workouts should be at least 20 percent below the upper Zone 1 heart rate.

ZONE 1: AEROBIC CONDITIONING

This intensity forms the base for aerobic development. Undervaluing it will undermine your aerobic development. Throughout your training life, this forms the foundation upon which all higher-intensity training rests. The benefits of Zone 1 include increases in blood volume, mitochondrial mass, and capillary density and aerobic enzymes in the working muscles, all of which improve aerobic metabolism. Long workouts at this intensity also improve fat metabolism. Early on in a running program, it prepares the body for the heavy-impact loading, thus helping with injury prevention. Blood lactate concentrations for this intensity should range between less than 1 to around 1.5mMol/L. Heart rate at the top of Zone 1 will be about 10 percent below your AeT (top of Zone 2).

Aerobic Threshold Speed Is the Single Most Important Measure of a Person's Aerobic System

Without this speed being as high as possible, no amount of high-intensity work will ever give the best endurance results. Kilian understands this concept well and that even in a year with a very full race calendar his training is still dominated by low-intensity aerobic base training. Understanding this fact is critical to maximizing your long-term results.

Recall from the Physiology Section that this unglamorous aerobic base is what allows you to sustain higher speeds for longer times. World-class endurance athletes across many sports typically spend about 80-85 percent of their training volume in Z1 and Z2.

NOTES FROM KÍLIAN

In 2016, I was training a bit more than 1,300 hours. Of this, I had approximately 275 hours spent at a high intensity, mostly Zones 3 and 4, and a bit of Zone 5, almost all while racing. I did a lot of racing that year. While training, I had another 350 hours pushing a bit, comfortable but fast aerobic, Zone 3. The other 675–700 hours were all much easier at an aerobic capacity-building intensity, Zones 1 and 2. This

year represented a high proportion of higher-intensity training mostly because I had a heavy race schedule. Note: this breaks out as roughly 20 percent racing, 26 percent at Zone 3, and 54 percent at Zones 1 and 2. Of course, in my younger years, especially as a teenager, I was not racing as much so my training distribution was more heavily weighted to base aerobic capacity work.

Duration of Zone 1 work should be a minimum of thirty minutes of continuous activity and can range up to several hours. Due to the duration, training at this intensity can be a significant aid to technique and economy training. It should account for between 50 and 70 percent of the annual aerobic training volume for all mountain athletes.

ZONE 2: AEROBIC CAPACITY

We frequently refer to the importance of the aerobic base when speaking of endurance sports; this is the key to endurance success. Zone 2 intensity requires the maximum output of the fat/oxidative (aerobic) metabolism. It has the greatest effect in raising the AeT, which is one of the most important benchmarks of aerobic development. For all but the elite-level athlete, this may be the most trainable aerobic quality. Raising it will provide the support needed to maximally improve endurance as measured by speed at the LT.

The benefits from Zone 2 training are much as they are for Zone 1 except that they accrue to a larger pool of muscle fibers due to the increased power output and hence greater muscle-fiber recruitment. Training is performed in a continuous mode with durations usually between 30 and 120 minutes. It can and should be utilized mainly through the base phase of training and plays a smaller role once higher-intensity training begins to be included in the plan as competitions near.

Those just starting aerobic training and/or those with Aerobic Deficiency Syndrome (see pages 46/47) can do almost all their aerobic base training in Zone 2 (foregoing Zone 1 completely) because their pace at this intensity is going to be slow enough that their muscles will not get too much of a beating. However, those athletes with a high AeT and whose pace at this threshold is fast will not be able to do much Zone 2 training in a week without risking overtraining. They will need to do much more of their aerobic base work in Zone 1 and even the Recovery Zone. Workouts in Zone 2 for them will be too taxing to do every day and need

to be limited to one or two days per week with a total volume of less than 10 to 15 percent of the weekly volume spent in Zone 2.

ZONE 3: ENDURANCE

While this is considered aerobic training, it is of a different nature than Zones 1 and 2. At this intensity, the increase in speed requires the muscles to up their ATP turnover. Glycolysis becomes the dominant source for ATP, both in the ST muscle fibers and as more FT fibers get recruited. Lactate levels begin to rise with the increased pace. Lactate accumulation remains at levels sustainable for up to an hour in well-trained athletes with no performance degradation. Because of the high energy demands to sustain these speeds and the duration that can be sustained at this intensity, both the aerobic and glycolytic systems get a hefty training stimulus from Zone 3 training. For the less well trained, this intensity will have a strong Muscular Endurance effect because the limit to this endurance is imposed by the aerobic capacity of those FT fibers needed to sustain this output.

The upper end of this intensity zone is defined by the LT (the speed above which lactate begins to accumulate faster than it can be removed). Beyond the LT, every athlete is operating on borrowed time and will be forced to slow the pace. Most athletes very quickly develop an intuitive sense for what this feels like.

Many people see Zone 3 training as the magic intensity, the one-stop-shopping of endurance training. They assume that since this zone has such a powerful

The Ten Percent Test

Zone 3 training is most effective when the athlete has sufficient basic aerobic capacity; when the vacuum cleaner is big and powerful. How much capacity is sufficient? When you've raised your AeT to be within 10 percent (elite athletes can have a Z3 spread of 6–7 percent or only 10 beats) of your LT as measured by either heart rate or pace. With more than a 10 percent spread between thresholds, an athlete still has aerobic deficiency and needs to build more aerobic base. Those who have less than a 10 percent spread between thresholds will need to reduce or even drop Z2 training, substituting Z3 workouts.

Here's how to do the 10 percent test: Determine your AeT using one of the methods described on pages 152 to 155

(AeT Testing). Then do the LT test (see page 155).

Calculate the percentage difference between the AeT heart rate and the LT heart rate by dividing the higher heart rate by the lower heart rate. We know this is not the conventional way to calculate percentage, but it works well for our purposes.

Example: Suppose your AeT heart rate is 128 as determined by a laboratory test. Your LT hill-climb test shows an average heart rate of 150. $150/128 = 1.17$. This shows that the LT heart rate is 17 percent greater than the AeT heart rate.

You still have a lot of potential to improve your aerobic base with Z1–2 and should not be too eager to move to adding Z3 or higher intensity yet.

training effect, it can be used almost exclusively to make endurance training more time efficient. Tempting as this may be we strongly caution you to not overuse this intensity because it puts you into the Utilization Training modality, with all the associated pitfalls. Zone 3 training is seductive because it will give an immediate boost to anyone. This intensity should typically make up no more than 10 percent of the total annual training volume. For those with insufficient aerobic base development, as indicated by the Ten Percent Test (see sidebar), overuse of this intensity can easily lead to a performance decline.

ZONE 4: AEROBIC POWER

As with Zone 3 training, this is another intensity that can elicit seductively rapid gains in fitness and is a necessary component in all endurance programs. This intensity above the LT maximally utilizes both the aerobic and the glycolytic metabolism of the working muscle cells as well as the lactate removal (shuttle) mechanisms. Most of the adaptations from this training accrue quickly. You feel yourself getting noticeably fitter from workout to workout. What's not to like about such rapid gains, right? Read on.

Polarized Training

In a recent study by Stephen Seiler (see note below) that looked at how world-class endurance athletes actually trained across several sports—cross-country skiing, running, cycling, and rowing—he found just the opposite of what the popular press advises. He looked at the actual training logs of many of the world's most accomplished endurance athletes, including multiple Olympic gold medalists, world champions, and Tour de France riders. He found a striking similarity across sports when he examined how much of their training volume consisted of training below the AeT, between the AeT and the LT, and above the LT. His collation of the data from these elites who, as professionals, must choose the best training methods to remain competitive, gave a clear picture of the preferred training intensity distribution. The conclusion he drew was

that they tended to train 80 percent of time below the AeT, roughly 6 to 8 percent between AeT and LT, and 12 to 14 percent at the higher intensity above the LT. This may seem highly counterintuitive given the benefits accrued from Z3. As we mentioned in chapter two, Seiler coined the term *Polarized Training* to describe this system. Another study (see below) by Seiler has shown performance gains using Polarized Training even with recreational runners.

Notes:

"What is the best practice for training intensity and duration distribution in endurance athletes?" Seiler. *International Journal of Sports Physiology and Performance*, 2010 Vol 5.

"Does Polarized Training improve performance in recreational runners?" Seiler. *International Journal of Sports Physiology and Performance*, 2014, Vol 9.

ATHLETE STORIES

The Message of Running

Rickey Gates

Historically, running wasn't about individual accomplishment and the spectacle of winning and losing. It was a means of communication. In 490 BC, before the Battle of Marathon, Pheidippides ran 125-plus miles from Athens to Sparta to enlist help against the coming Persian forces. When the Spartans delayed, he ran back to Athens with the bad news. He wasn't the first to carry crucial information over an ultradistance. The original runners in ancient cultures around the world, from Central and South America to Japan, were messengers.

Today, I see being a messenger as a more valid goal than being a champion. But I don't pass judgment on those who are single-mindedly focused on a race or on a world record, because I went through that same journey before reaching my current philosophy. Early in my running life, I chased the spectacle.

When I was in college, at the University of Colorado, I decided I wanted to make their highly competitive cross-country team. I'd just taken two years off school to travel, and now I fixated on this goal. My training was tightly regimented: I timed and measured every single run. I didn't dare step out the door without a watch. For three Augusts in a row I lined up to race against the team, hoping to finish within a minute of the leaders. I came up short each year.

After my third and final attempt, I realized I'd been training to be a runner I wasn't. Once I got back into the mountains, I saw that I'd been pursuing a goal that was not *my* goal. It wasn't who I was. My true love and appreciation for the sport took off thanks to that failure.

I ditched my watch and I stopped keeping track of miles. I settled on a new objective: making the US Mountain Running Team. To that end, I still needed

some sort of program, but instead of being overly strict about it, I decided to run by feel. My plan, which I followed for several years, was simple: go for one run a day. On a day where I had absolutely no interest whatsoever in going for a run, maybe I'd turn around at the base of the driveway. According to my rule, that counted. But most of the time, even if I didn't want to go for a run and yet still made the effort of putting on my shorts and my shoes and heading out the door, I would go much farther than the end of the driveway.

Sometimes we can feel really good, but our minds aren't into it, and sometimes our minds are into it, but our bodies aren't. It's important to keep the mental and the physical in balance and to listen to both.

Based in part on this tuned-in approach, I made the US Mountain Running Team when I was twenty-four. The year after that, I won two national titles: US Mountain Running Champion and US Trail Running Champion. Running in the mountains is me; that's who I am.

Not only did trail and mountain running bring me a newfound appreciation for how I can interact with the environment, it also became a way for me to indulge my passion for traveling. As racing became increasingly less important to me, I embraced running as an opportunity for exploration—at home and abroad. I came to see it as the ideal vehicle for immersing myself in new places and meeting new people.

In 2017, I embarked on a true cross-country adventure: from March 1 to August 1, I traced a 3,700-mile route from South Carolina to San Francisco. Like those ancient runners, I wanted to be a messenger—to use running as a medium of connection and communication. There was no advance training involved; I didn't do much beyond putting on as much weight as I could, because invariably I was going to lose it all (which I



did, plus some). Along with a lightweight pack (and a baby jogger while crossing the desert), I carried a clear message from one state to the next: we may not agree politically, and our backgrounds may be divergent, but I'm a good person and you're a good person and we have more things in common than we do differences. I barely noticed when I crossed state lines.

After stitching that thread of commonality from coast to coast, I hatched a follow-up adventure: Every Single Street. Over forty days in the fall of 2018, I'll run every street in San Francisco—all 1,200 miles. My run across the country was a very long, thin line, and running every street in San Francisco will be like grabbing a crayon and vigorously coloring in this urban grid until there's no white left. I experienced how diverse the country is, and I suspect that a city—a seven-by-seven miles square, the size of San Francisco—can be equally diverse, as long as I allow myself to be curious and open to new people, places, and opportunities.

If there's one word that could be used to describe my running over the years, from my aspiring cross-country days to now, it would be *consistency*. I no longer run every day, but I run often, because without maintaining consistency, I can't mine the depths of this sport.

The very act of running has been my messenger: over my fifteen-year evolution as a runner, running has unlocked its own potential—a potential that lies beyond races and competition. It's not about going faster; it's about slowing down, opening my mind, and engaging with what I see and hear. It's about being a modern-day messenger.

Rickey Gates is piecing together a visual and written journey of his run across America and learning that the journey is actually easier than the telling of it. He lives in Oakland, California, with his girlfriend and motorcycle, the Freedom Machine.

Rickey Gates runs through the desert near Green River, Utah, on day 113 of a five-month run across the United States. He picked up this golf caddy in Moab to help him transport the amount of water necessary to cover ninety miles of desert at a time. Photo: Rickey Gates



Zone 4 workouts, like Zone 3, also have a very powerful muscular endurance training effect. Especially for the less well trained, the duration of Zones 4 and 5 workouts will be limited by a lack of localized muscular endurance. You'll recognize this if your legs are screaming at you but your heart rate tops out in Zone 3 levels. When this happens, you are not getting the intended benefit of a Zone 4 workout. You need to improve your muscular endurance before you will be able to sustain the high heart rates of Zone 4. For more on muscular endurance, see the Strength Section.

ZONE 5: MAXIMUM ANAEROBIC EFFORT

This is maximum intensity and can be maintained for less than thirty to forty-five seconds. Due to the lag in cardiac response, heart rate is not a good measure of this workload. This intensity requires that your brain recruit the bulk of the pool of available motor units available for the task. This intensity taxes the glycolytic metabolic system and the neuromuscular system and is a great stimulus for improving sport-specific strength and technique, and finally it is the best way to develop anaerobic power endurance critical for short events like the skimo sprint.

Zone 5 training can and should be included in most training periods for endurance athletes. Early in the training cycle it will consist entirely of ten-second, very powerful sprints, usually up a steep hill, and only later will the duration be extended to produce an endurance effect.

The downside to this sort of training is potential injury due to the high loads imposed. A good strength base will help prepare you for this powerful work. It should always be done in interval fashion with very long rest intervals between work bouts. To maximize the power in each repetition, you need to allow enough time for the muscles' ATP stores to replenish. Three minutes is sufficient to restore most of the ATP. If you notice a significant drop in power after a few reps and longer rests don't help, you should conclude the workout. You have fatigued the FT motor units and done the job you intended to do. These workouts can be mentally tough for endurance athletes who think resting is counterproductive and get antsy during the rest intervals.

Continuity, Gradualness, and Modulation

These three words are the principles of all successful training programs. Keep these in mind as you build your program to prevent you from making gross errors.

Correct training places one or more of your body's systems into a crisis state. These stresses impact the body's structural and functional systems. The myriad adaptation processes that occur to both these systems are induced by varying stimuli and take place at varying rates.

A failure to adapt (known as stagnation) can occur or even worse can cause overtraining, which is a diminishing level of fitness when these key elements are



Jeff 'Bronco Billy' Browning sees his workout ending sooner than planned as he gets chased out of the alpine by an approaching thunderstorm. San Juan Mountains, Colorado. Photo: Steven Gnam

ignored. It takes time for the body to adjust to the training load you impose. A reapplication of another similar training stress will not give you the benefits you seek if your body's structural and functional systems have not sufficiently adapted to your current training load.

CONTINUITY

Continuity in training refers to maintaining a regular schedule of training with minimal interruption. You have to be motivated and disciplined to fulfill the requirements of the plan you undertake. Obviously, there needs to be some flexibility to account for the unexpected. Lapses in training do happen. They can be managed, but they cannot be overlooked. If you miss a week of training due to work, travel, or sickness, you can't pretend that you actually did all that training and progress to the next part of the plan. Your body will not be prepared for the next step, and you'll likely have setbacks.

How you manage this discontinuity depends on its length and the reason for it. Occasional breaks of a day or two are not much of a problem as long as they do not diminish the training load by more than roughly 5 percent in a month. Frequent or prolonged breaks from training that amount to more than 5 percent of the monthly volume do create a need to adjust your subsequent training by taking a step back in the plan.

GRADUALNESS

The most effective training must be progressive, meaning that the training loads need to *gradually* become larger than those that have come before in order to have a beneficial effect. As your fitness increases, what used to be too much will gradually become not enough.

Gradualness is an acknowledgment of the time your body needs to adapt to the training stimulus. Given the right training stimulus with adequate recovery, your body has a remarkable ability to adapt to frequent gentle nudges. But it does not respond well to infrequent bludgeoning. Occasional bursts (or lapses) of enthusiasm and motivation that result in big changes in training load are a common approach to training by beginners and dilettantes. The results from this sporadic exercise (we can't call it training) will never lead to optimal results and can be disastrous. Gradualness is a virtue that cannot be overemphasized.

Once your body is adapted to the stress of the current training load, what used to create a physiological mini-crisis will no longer be enough load to cause your body to keep adapting. Some creative changes to the training load need to be implemented for the next cycle or progress will stop as you become adapted to this old level of stress. The types of changes vary according to the qualities you seek to develop and tend to fall into two distinct categories: volume and intensity. For a quality such as aerobic capacity, these adjustments are simple and involve mostly an increase in training volume. To effect simultaneous adaptations of different qualities like anaerobic capacity and speed, or aerobic endurance and muscular endurance, more elaborate training methods need to be implemented and must be administered with care and finesse.

A gradual progression of the training load means different things for different athletes. Beginners and those with a low annual volume of training can progress faster than those with many years of training who are closer to their ultimate potential fitness. As a general rule, beginners (under 350–400 hours each year) can increase training load, as measured by time, by as much as 25 percent per year. Advanced athletes, training more than 500 hours per year, will be unable to make such jumps because they are already operating much closer to their limit. For them, a 10 percent jump in yearly volume should be considered a maximum. Elite-level athletes often make no change in overall annual training volume, instead seeking gains by juggling the type and amount of higher-intensity and specific training.

MODULATION

Modulation refers to the fluctuating level of the training load that allows the body a chance to recover its homeostasis after a build-up period. Modulation allows you to stress the targeted systems, then give them time to adapt before applying a new training load. Depending on the systems you target, this modulation can follow a cycle measured in hours, up to one that's measured in weeks. With care,

an athlete can modulate training to allow several systems to adapt at the same time. Being mindful of and recording your daily perceptions of how you feel before, during, and after training in a training log helps you learn how you adapt to the various stressors in your life.

An extreme form of modulation is called overreaching; it is sometimes employed by advanced athletes. This entails a short, planned period of very high training loads preceded by a slightly below average week and followed by a substantial recovery period. Overreaching could be two massive back-to-back weekend workouts followed by three of four light days. Or it could be an entire week staying high in a hut and packing in long days with a couple of interval workouts that might necessitate a full week for recovery.

Specificity and Cross-Training

By specificity, we mean training in ways that look like the event being trained for. After all, elite-level swimmers principally swim for training, champion cyclists ride, top runners run, and world-class skiers (snow dependent) ski. There is a good reason for this. The better the athlete, the more event specific the training must be to see appreciable gains.

That being said, beginner endurance athletes and those with long lapses in their training history can make initial gains in fitness by doing almost any prolonged endurance activity. In the lexicon of our times, this nonspecific training is referred to as cross-training.

As your fitness improves, your gains in fitness will become more elusive unless you shift to activities directly related to your intended activity. For example, at this stage in his career, Kilian won't see any benefit to his ascent speed by swimming laps or paddling a kayak (but these may be beneficial recovery workouts). He must train in ways that mimic the demands of his intended events.

As an uphill athlete (even a beginner) seeking to improve your endurance, you should give priority to weight-bearing exercise. Cycling is a great general exercise. However, bikes are just too efficient and this makes them less effective training tools from the time and specificity standpoint. The sitting position on the bike means that you do not have to support your full body weight, which greatly reduces the energy cost of the exercise and the muscle mass used to propel yourself. The bike also limits the range of motion, eliminating the coordination, balance, and variability of the footing required while covering uneven terrain. Cycling can be great for recovery workouts and when rehabilitating from injuries. Kilian used to use cycling in a high gear as muscular endurance training. But, it formed a very small part of his training volume and now has been dropped.

Swimming is another great exercise that has little carryover to our mountain sports but is an excellent recovery tool. The prone position in swimming means

the heart doesn't work as hard to pump the blood. The water also keeps the body cool, leaving more blood available for the working muscles. If this is beginning to sound like running and hiking should be emphasized in your basic preparation time, then you are getting the picture.

What are the implications for the nonelite-level athlete? Any exercise, regardless of modality or intensity, will benefit an unfit person. But the benefits of cross-training diminish pretty fast once you are at the competitive recreational level. If you can only handle running less than twenty-five miles (forty kilometers) or six hours per week, then you will see benefits from adding other modes like hiking, cycling, rowing, stair machine, even swimming. If, however, you are comfortable with a much higher training volume and want to maximize performance gains, then your additional training energy needs to be devoted to running. The sports covered in this book are all foot borne and require adaptations that will only occur from foot-borne training.

Individuality

“If you give ten different athletes the same workout, you will see ten different responses.”

– Renato Canova

The individual nature of each athlete's response to training has made many a coach's hair turn gray. This aspect is often either overlooked or misunderstood in the application of training methods. Factors that influence individuality are:

- **Lifetime training history.** Were you inactive during much of your life until recently becoming interested in endurance sports? Or were you part of a running or cross-country ski team through adolescence? People with years of endurance training, especially in childhood through adolescence, will have retained many of the structural adaptations made during those years even if they have been relatively inactive recently. Adaptations to training come much slower after the teens and twenties and will require much more work to make happen after middle age.
- **Recent training history.** Have you been hitting the gym religiously three days a week for a CrossFit burn for the past two years? Or have you been logging six to ten hours a week of running and hiking in the mountains? Or have you been sedentary the past ten years due to work and family constraints? All these will impact how you should approach your training.
- **Life stressors.** The variety of stress in your life from school, work, family, and relationships all will impact your ability to handle training loads.

- **Fast- or Slow-Twitch muscles.** Are your muscle-fiber types naturally FT or ST dominant? This too plays a role in how you respond to endurance training. See the sidebar below.

If we all responded like a machine, the same way to every training input regardless of the above factors, then training would be as simple as following a road map from point A to point B. But we are highly complex organisms and each of us has unique qualities we bring to this venture that result in an individualized, unique response. One of the most common mistakes made is to copy the training of someone else, especially when that person is very successful in his or her sport. The popular media loves to report the details of the final six weeks of workouts that led to so and so putting in a record-breaking performance. While this can be inspiring, it is all too easy for the readers/viewers to believe that if they too complete these magical workouts, they too will have extraordinary results. If these journalists really wanted to be helpful they would be telling us what those champions did for the past ten years that allowed them to not only handle those final workouts, but to benefit from them. It is the long-term, capacity-building training history that determines the effectiveness of the short-term utilization workouts.

Are You a Fast-Twitch or Slow-Twitch Athlete?

Without going to the pain or expense of a needle biopsy, there are some other ways to determine whether you are FT or ST dominant, including the following.

FT Dominant:

- Good sprint performance
- Good vertical leap
- Produce high levels of lactate during easy workouts
- Can't hold peak fitness for long

ST Dominant:

- Do not have a fast sprint
- Poor vertical leap
- Cannot produce much lactate at high intensities (low anaerobic capacity)
- Maintain peak form for longer
- Recover from long, hard aerobic training quickly
- Do not need as much fueling during long workouts

FT Needs

- More aerobic base (Z1-Z2) training to become less reliant on anaerobic system
- Should not use much long Z4 training. Better with shorter repetitions (<2 min) to keep lactate low

ST Needs

- Frequent, small doses of speed to improve anaerobic capacity

Training Stimulus and Recovery: How They Are Related

NOTES FROM KÍLIAN

I was already doing many days of eight to ten hours of road cycling when I was twelve. My idea then was to do long days cycling, pushing a big gear because I wanted to be a strong climber. I started serious training when I was thirteen when I started training properly with a coach; that first year I trained 600 hours. I discovered that I like to train and suffer. We did many long hikes in the Pyrenees. I was building a lot of endurance when I was young. Starting that year, I did 80-kilometer (50-mile) runs; long trainings. I was very young for such long runs.

And then I kept going, adding about 10 percent more training hours a year. For the last eight years I have held my annual training volume between 1,000 and 1,300 hours. If you want to increase to this high number of training hours, you have to increase no more than 10 percent a year for several years.

I know some world-class skiers and mountaineers who train 600 hours a year and they are very good. For shorter-distance running and ski-mountaineering races, this is sufficient. For the really long races, the higher volumes are necessary.

From earlier discussions, you know that your body's initial reaction to the stress of training is to become weaker. After sufficient recovery you achieve a higher level of fitness than before the training bout. But, just what do you do in this recovery period? Is it bed rest? Is it reduced volume or intensity? How long do you need to recover before the next workout?

These important considerations dictate that correct training is a balancing act. The writing of a training plan is the easy part. The tricky part is the successful implementation of that plan so that you arrive at your highest level of fitness at the correct time. This is where many athletes get it wrong and what we hope to help you avoid.

The most underestimated component of a training plan is the recovery. Some level of fatigue will be your nearly constant companion when training. Fatigue is your body's feedback to you about the training. Learning how to interpret that feedback and respond appropriately will be fundamental to successful training.

RECOVERY TIMES

Let's suppose that you are tired from the previous day's exertions. Given that it takes between eight and seventy-two hours to restore intramuscular glycogen, it is usually enough to just delay the next heavy training bout for a day or two. When fatigue is significant enough to prevent another similar or harder workout than Zone 1 or Recovery intensity, aerobic training must be done on the intervening days before the next similarly challenging workout will be effective.

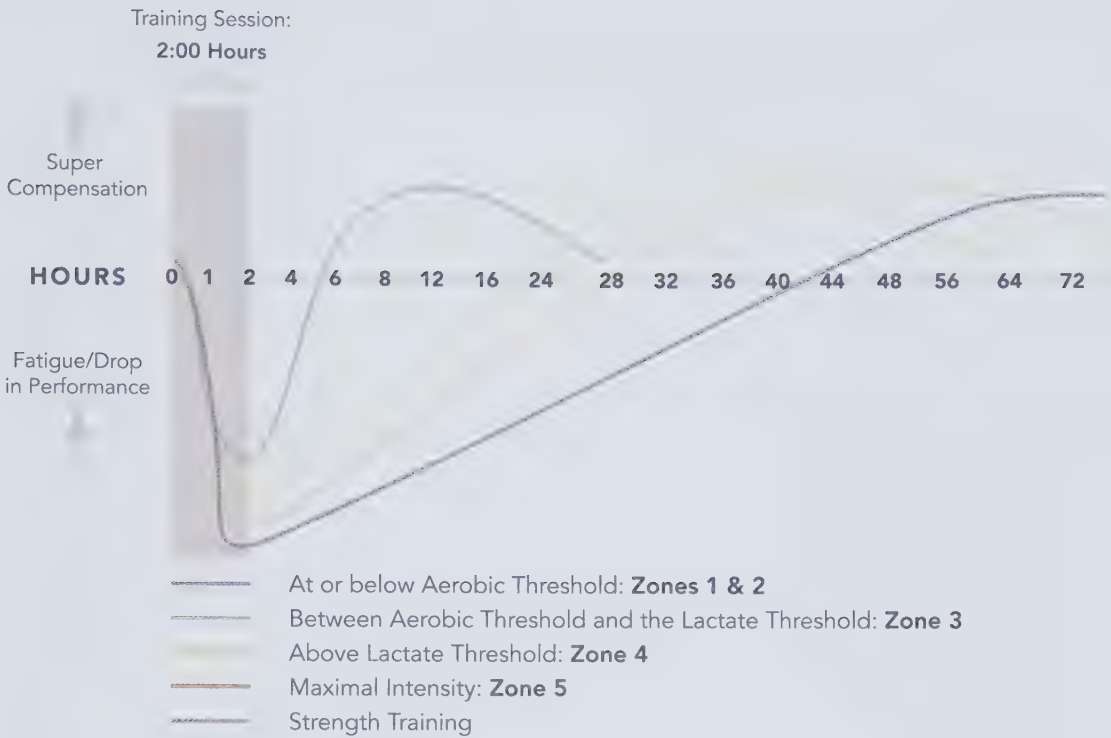


Fig. 3.4 Recovery and Adaptation Times

The vertical scale is purely qualitative but the horizontal scale is in hours. While there is some interpersonal variability to these numbers, the originator of this scale collected data on hundreds of swimmers over many years to arrive at these values. It has been the authors' experience that these values hold quite well. Reproduced from *The Science of Winning* by Jan Olbrecht, courtesy of F&G Partners Publishing.

Endurance training has a dramatic effect on the amount of the intramuscular energy stores, with better-trained athletes having as much as three times the intramuscular glycogen stores of the untrained person. That means that these recovery times between workouts are highly individualized. However, this is somewhat balanced out by the fact that the better trained will usually be training harder and for longer, so in the end, rough guidelines do present themselves.

High-intensity exercise and strength training require the longest recovery times of two to three full days before you'll be prepared to absorb a similar load. Low-intensity endurance workouts of two to three hours or less will normally be recovered from in eight to twenty-four hours. Long-duration (over two hours) training at even low intensity may require twenty-four to forty-eight hours for the restoration of glycogen stores. Refer to figure 3.4. Eating during and within thirty to sixty minutes of finishing these long workouts will greatly reduce the time needed for recovery, sometimes by days.

ATHLETE STORIES

In the Mountains, Durability Equals Speed

Mike Foote

When I first started running 100-mile ultramarathons in 2009, my primary goal was to be fast. However, after a decade of racing and adventuring in the mountains, I've realized that durability, not speed, should be the North Star of my training.

The events that appeal most to me, steep and slow 100-mile mountain runs like UTMB or the Hardrock 100, are races of attrition. People get injured, people slow down, and people DNF. Although it is contrary to think about it this way, the people who finish first aren't necessarily the fastest; they are the ones who are the least broken and who slow down less overall. I grew more attracted to this mindset rather than one of pure speed. Instead of striving for speed, I limit slowing down. To achieve that, I cultivate durability. And for years, this desire for durability has informed how I train.

In 2015, alongside a couple of close friends, I ran from my front doorstep in Missoula, Montana, to Banff, Canada, crossing the region known as the Crown of the Continent. Over three weeks we covered 600 miles of mostly off-trail ridgelines, traversing thirteen mountain ranges in the effort. Day in and day out we soldiered north through technical terrain, chossy climbs, and heinous bushwhacks. Throughout the expedition, I was pushed to my limit, but I never questioned my ability to finish, and I didn't have a single injury flare up. Looking back, I don't think I would have realized that dream if speed was all I cared about. My durability got me through.

The bad news is that there are no shortcuts to durability. The good news is that working toward making both your body and mind more resilient to the stresses of training and racing—and dare I say life—is a fun and engaging process.

A few key components of my training that have increased my durability are training volume, strength work, and prioritized self-care.

I'm a firm believer that having a high-volume training load will set you up well for being able to handle the workload of big races. The downside is that it can take time to build the capacity for such big goals. But it's worth it for having that well to dig from twenty-two hours into a twenty-four-hour push.

Also, I believe in the positive benefits of strength training. Going to the gym, even once a week, can have a huge impact on your ability to stay injury free and moving forward.

Finally, I focus on all the ancillary work that supports the training. I prioritize massage, foam rolling, yoga, and more in order to keep my body "put together." Also—spoiler alert—I've learned that diet and sleep are important as well. Shocking, I know.

Looking back on the last ten years of training for and racing ultras, I don't think I'm any faster at running a mile. But I seem to be running 100 miles in the mountains quicker than when I began. And that's good enough for me. ■

Mike Foote is an aspiring efficient mountain traveler who has stood on the podium at many of the biggest ultramarathons in the world. He is also a member of the US Ski Mountaineering Team. In 2018, Mike, coached by Scott Johnston, climbed and skied 61,200 feet in twenty-four hours, setting a new world record in the process.

Mike Foote refueling at the Chapman Aid Station, about 80 miles into the 2017 Hardrock 100. Mike placed second behind Kilian Jornet in that race. Photo: Steven Gnam



NOTES FROM KÍLIAN

I have never needed to think too much about special recovery techniques because I can recharge quickly. I know that muscular fatigue from hard efforts like steep uphill intervals and the recovery from long aerobic training require different kinds of recovery.

Fatigue caused by hard muscular endurance work can take seventy-two hours to recover. If I do something like a fast race or a day in the mountains with a lot of vertical, and destroy the muscles, that will take me two days to recover from. I can keep going the next day but I will have heavy legs and I go more slowly on those days.

Quality of rest is also very important. Staying at home, working around the

house, sitting in the office—this is not proper resting. Rest means doing nothing or doing everything very slowly.

Aerobic fatigue is different. I can do things for a long duration, but I find that to determine the right amount of recovery I have to listen to how my body feels. If I feel the training will not do anything for me, that it will not improve me, I stop and take a rest day. If every day is like that for you, maybe you are not well trained enough for the level you are pushing. And some days you might feel like you are doing the workout just to do it; at that point it might be important to rest.

If you do not feel a progressive increase in energy, strength, and overall fitness from week to week, you are either training too hard, not allowing enough time for recovery, or your active recovery sessions are too intense. There are always a few “blah” days here and there, but if the general trend for a week is down, not up, you need to address the situation. The first step of acknowledging the fatigue is usually the toughest one.

Far too many times, we have had athletes ask us training questions after they have become injured or overtrained. And invariably, when we explain these concepts, their response is, “Oh, I guess I was definitely overtraining or underrecovering.” By the time they ask, “Now what do I do?” it is often too late for all but the most drastic measures.

ENHANCING RECOVERY

Certain steps can and should be taken to speed up the recovery process.

For athletes who are pushing the limits of training that their bodies can absorb, recovery strategies become an integral part of the overall plan. For those who are time limited rather than energy limited in the amount of training they can fit in during a given week, recovery can still be a useful tool, but may not be so essential.

Luke Nelson, a little haggard after running the 2014 Rut 50K on Lone Peak, Big Sky, Montana. The 2014 race drew the top mountain, ultra, and trail runners from around the world as it served as the final race in the Sky Running series. Photo: Steven Gnam





Christian Varesco loosens up with a run in view of the Pale di San Martino range clearing after a summer storm. Dolomites, Italy. Photo: Federico Modica

The following is a list in order of importance of some of the most common methods used to enhance recovery.

Diet

This is most the important component for recovery.

Food is fuel. And you need to eat with an eye toward recovery. What does this mean? Realize that heavy and prolonged exercise has a depleting effect on your nutritional stores, especially glycogen. The sooner you can begin to get the caloric building blocks into your system, the sooner they can start to repair the damage and depletion done by the training bout. While research indicates that there are some variations in effective post-exercise feeding strategies, some general guidelines seem to arise.

Consume 100–200 calories in the first thirty minutes post exercise of over an hour. Have these calories in a form that does not upset your stomach (you may need to experiment). Eat these in a ratio of 3:1 or 4:1 of carbohydrates to protein. It is during this post-exercise window that the muscles can replenish their glycogen stores quickly. Miss this window and recovery can be delayed by up to days depending on the workout/race.

Sleep

We all have a general sense of how a good night's sleep or lack of affects our energy and outlook the next day. There is solid science to back up your seat-of-the-pants feeling.

Quality sleep is vital for the recovery process. A lot of important stuff happens during REM (rapid eye movement) sleep that aids your body in repairing damage inflicted by training. Much of that rebuilding comes about as a result of various hormones released during sleep. REM sleep is a critical phase of sleep that is not definitively understood, but we do know that it occurs in several blocks of 90 to 120 minutes throughout the night and it involves a state of very deep relaxation that aids the recovery process. Two of the best stimulators for the release of your body's natural human growth hormone, known as somatotropin or HGH, are REM sleep and vigorous exercise. HGH is one of the body's primary tools for adapting to higher training loads. Enhancing its production is an easy and smart way to develop better strength and endurance.

Anything that gets in the way of REM sleep is going to have a serious impact on your recovery. Ever spend a fitful, sleepless night? How recovered did you feel the next morning when it was time to tackle a big day? But getting even fifteen minutes of deep sleep during a nap can have an amazingly refreshing effect.

Massage

Stiff and tight muscles are often the most obvious sign that you have been working your body hard. Endurance activities involve many thousands of repetitions of the same or very similar movements. This repetition often results in acute inflammation of these muscles, which you perceive as stiffness and soreness.



Prior to the 2015 Mont Blanc Marathon, Kilian Jornet receives a painful massage from his physiotherapist, Arnaud Tortel. Chamonix, France. Photo: Jordi Saragossa



Kilian Jornet using his Compex electronic muscle stimulator machine to help his legs recover between training days during a high-volume training week. Chamonix, France. Photo: Jordi Saragossa

The inflammation actually stimulates some of the adaptations to the training, but it can be too much of a good thing. Massage by a skilled practitioner increases the blood circulation in tired muscles. Improved circulation speeds the replenishment of vital nutrients, reduces this muscular inflammation, and accelerates recovery. This is the reason professional cyclists use massage after heavy training and between stages on multiday races. It works! A recent study by Dr. Mark Tarnopolsky of McMaster University showed that as little as ten minutes of deep-tissue massage had a dramatic effect on speeding recovery.

Self-Massage

Since most of us can't access or afford a daily massage, it's nice to know that self-massage can be equally as effective. Numerous tools for self-massage have been developed and we've tried many but have come to rely on balls. They are small and easy to travel with. They allow you to get more directed pressure on small or larger areas than a foam roller can. Our favorites are the MobilityWOD 80mm ball and the Roll Model balls. The book *The Roll Model* by Jill Miller is a masterpiece on how to use the ball-rolling techniques to speed recovery and treat chronic musculoskeletal problems. We won't go deeply into this practice but highly recommend the book.

Basically the ball rolling produces a shearing effect on the muscle fibers, which reduces adhesions and loosens stiff muscles better than anything we have used outside of the hands of an expert sports massage therapist. You control the pressure and the positioning so that you target just the area you need to. For some



Athletes letting off some steam during the Vertical Crepa Neigra race in Canazei, Val di Fassa, Sass Pordoi, Italy. Photo: Federico Modica

people, the pressure of the ball will be too intense and a foam roller will be a better initial tool. Once you are able to apply full weight to the foam roller, it will be time to think about progressing to the balls. Using these self-massage techniques right before bedtime will help you relax and prepare you for sounder sleep.

Electro stim machines can be very effective recovery tools because they act a bit like massage when used in the recovery setting.

Reducing Life Stressors

Using relaxation techniques such as meditation can be an aid to recovery by reducing ancillary stress. Your body responds to stress, in large measure, with the same reactions regardless of whether that stress is a result of work or of training. If you have a stressful life, then it may be that one of the best training strategies for you is finding ways of reducing that life stress so you can handle more training stress.

Recovery Workouts

Calling these *workouts* is a bit of a misnomer; they should not be viewed in the same way as normal training. While certain ones can have a maintenance effect on aerobic qualities, their main use is to speed the recovery process. While it may



Laura Orgue playing in Arches National Park, Utah. Photo: Martina Valmassoi

seem counterintuitive that more exercise will do anything other than deplete your already-taxed reserves, these very-low-intensity sessions can have a powerful restorative effect. Coaches have known this for decades; the scientists are still trying to figure out why.

Keep in mind the following when using recovery workouts:

- The intensity must feel very easy. More discussion about this is found in “Understanding the Language of Intensity” in this section.
- They are most commonly used by athletes who are stacking workouts as closely together as possible, often twice a day, but can be helpful after an especially tough workout.
- Alternate modes of exercise such as cycling, paddling, and especially swimming will give needed relief to overused legs.
- Even a brisk walk can be a good recovery workout if you are really tired or feeling a little sick. Light exercise almost always beats being sedentary for aiding recovery.



Jeff Browning (no. 2), starting the 2013 San Diego 100 in which Jeff placed first with a time of 16:59:24. California. Photo: Jeff Johnson

ATHLETE STORIES

The Everest Marathon: My Wild Introduction to Ultrarunning

Anna Frost

There I was on the ground, only three kilometers from the finish of the 2009 Everest Marathon. I was crying, barely able to breathe, covered in dust and blood, and nursing a massively swollen ankle from earlier in the race. And now a yak had head-butted me. *A yak.*

I'd been in Nepal for eight weeks, and the yaks had never bothered me. But this one just didn't like me. It turned around and rammed into me, splitting all my knuckles open and knocking me to the ground. I was lucky that it hit my hand first, because otherwise it might have broken a rib.

The yak herder trotted over, picked me up, and pushed me along the trail. "Come on, come on, keep running, keep running!" he urged. I continued toward the finish, still crying.

Up until about a week before the marathon, the training had been going well. This was my first big non-European international race as a sponsored runner, and I'd arrived early to acclimatize. I also wanted to explore the area. I hired a porter to carry my bag from teahouse to teahouse so I could run the long way to the next village—up and over a 5,000-meter pass and back down through another valley. My porter would tell me before I left which hut he would be at, and I'd meet him there. It was an amazing way of being liberated to go longer and lighter.

Each time I reached a high pass, I felt really small. There you are at 5,000 meters, and it's still 3,000 more meters up to the tops of the highest mountains around you. It makes you realize how insignificant you are, and it reminds you to be grateful for Mother Nature.

In addition to these point-to-point runs, I was also doing my typical mountain-running training. Two or three days each week, I would stay at the same teahouse and do hill repeats. I'd find a steep trail and do anywhere from 30-second to 5- or 8-minute

intervals. As I got more and more acclimatized, I was able to push harder and longer at those intervals.

More than once while on a run I was invited into a stranger's kitchen for a cup of hot coffee or tea and *momos*, little rice flour dumplings with vegetables inside. I wasn't permitted to pay for the goodies, probably because I was so skinny at that point I looked like I needed some fleshing out. But it's also the ethos of the place: the people are happy and generous, so willing to give anything they can, even when they don't have much themselves.

After six weeks of exploring, training, and acclimatizing on my own, it was time to meet up with the rest of the Everest marathoners for our group hike to Base Camp. During the trek, I got really sick with the Khumbu cough; between the altitude and the hacking, I could hardly sleep. The closer we got to the start line, the worse I felt.

Race morning was no exception, but I had to do it. Off we went, and about 10 kilometers in, I twisted my ankle, practically touching the whole side of my shin to the ground. It swelled up immediately. I popped some painkillers and anti-inflammatory meds, shoved some snow in my sock, and kept running until my ankle got numb. The only other way out of there would have been on a donkey, and I had no interest in finding and riding a donkey.

Even with my cough and my stiff, puffy ankle, I was having a great race. I was leading the Westerners, and there were just two Nepalese guys far, far ahead of me—so far ahead I wasn't worried about their position. I just wanted to finish.

At a turnaround point beside a nunnery at about five kilometers to go, the nuns all came running out when they saw me and put white ceremonial scarves—ribbons of blessing—around my neck. They were really inspired to see a woman racing.



Anna Frost trains near Qaanaaq, Greenland. Photo: Kelvin Trautman

My yak encounter came just two kilometers later, which led to the herder encouraging me onward, toward Namche Bazaar. I made it, and I don't think I've ever been more thrilled to finish a race. I consider it my first ultra; yes, it was a marathon, but it was an ultra to get there and get through it—and get through the following day.

I woke up the next morning in a lot of pain. My ankle was enormous, and it was black all the way up my leg. My cuts from when the yak hit me were bright red and full of pus, because the dust up there is teeming with bacteria and disintegrated yak and horse poop. I'd developed conjunctivitis in my eyes as well, also thanks to the dust. I had some tea tree oil, so I rubbed that in my eyes, on my hand, and on my ankle. Then, using two sticks I found in the forest for support, I hobbled and wobbled my way down to the airport in Lukla for my flight back to Kathmandu.

It took me twelve hours to cover those 17.7 kilometers—the last leg of my first ultra.

Despite all the discomfort and unexpected twists and turns, I fell in love with the Himalaya. This good,

bad, amazing, and fulfilling experience has inspired me to return year after year—three times to Nepal and now three times to Bhutan as well. I like to be reminded of the generosity and compassion of the people and that you don't need as much as you think. That less is more. It's true in life, and it's especially true in a race, when all you really need is the drive to keep putting one foot in front of the other toward the finish, even if one foot is swollen and there's a cantankerous yak in your path. ■

'Frosty' Frost is a New Zealander with many passions, including two that intertwine beautifully: running and traveling. She has been doing both since 2004, exploring and discovering the world and herself through the highs and lows of the mountains and life. After nearly a decade spent traveling internationally and racing constantly—and mostly successfully—across all distances and types of terrain, she hit physical bottom. But she has come back with new life in recent years, having learned that running is just running. It is what she does, not who she is.

